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1 Generating Environments

1.1 `generate_environment(int: number_of_obstacles) → environment`

Input: An integer `number_of_obstacles`, representing the number of obstacles to include in the environment.

Output: An `environment` list, where each element is a dictionary representing an obstacle with randomly generated dimensions and position in a 20x20 grid.

This function creates a list of obstacles, each with randomized dimensions $w, h \in [0.5, 2]$ and a random pose $q \in SE(2)$. Each obstacle is represented as a dictionary containing:

- **width:** obstacle width
- **height:** obstacle height
- **position:** a tuple representing the (x, y) position
- **orientation:** the obstacle's rotation in radians

The environment therefore is represented as a list of obstacle dictionaries.

1.2 `scene_to_file(environment: env, string: filename)`

Input: An `environment` list and a string `filename` for file saving.

Output: Saves the environment configuration to a text file in `env_files/`, with each obstacle's attributes on a new line.

This function iterates over each obstacle in the `environment` list and saves it to a file in the format:

```
width height x y orientation
```

1.3 `scene_from_file(string: filename) → environment`

Input: A string `filename` that specifies the environment file to load.

Output: Returns an `environment` list with obstacles represented by dictionaries with width, height, position, and orientation.

This function reads a specified text file from the `env_files/filename` and reconstructs each obstacle from the saved parameters.

1.4 `visualize_scene(environment: env)`

Input: An `environment` list.

Output: A visualization of the environment with obstacles plotted as rectangles patches.

The function plots the environment using `matplotlib`, displaying each obstacle at its specified position, orientation, and dimensions within the 20×20 grid.

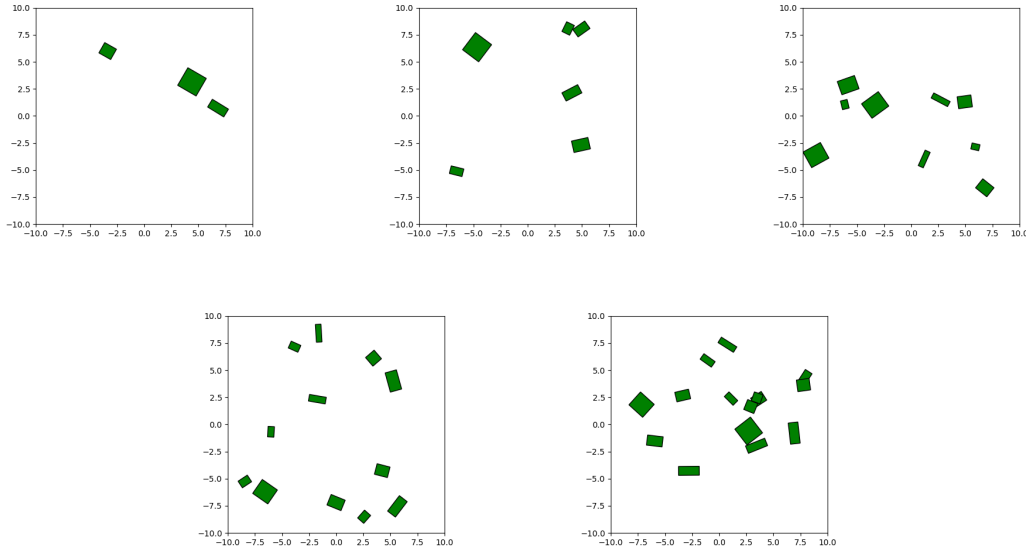


Figure 1: Five Environments

2 Nearest Neighbors Search (5%)

We implemented the nearest neighbor functionality as a linear search through all neighbors to identify k nearest neighbors. The code takes in command-line arguments specifying robot type, target configuration, number of neighbors ($-k$), and a file with configuration data ($--configs$).

2.1 Functionality

Inputs:

- **--robot:** Specifies the robot type, either "arm" or "freeBody".
- **--target:** Defines the target configuration. For "arm" robots, this consists of two joint angles (θ_0, θ_1) . For "freeBody" robots, it includes (x, y, θ) , where θ represents orientation.
- **-k:** An integer specifying the number of nearest neighbors to output.
- **--configs:** Filename containing M configurations, each on a new line with N values. The config file is an environment file generated from Part 1

Output: A visualization of the k nearest neighbors and the target configuration, with $k + 1$ robots shown.

2.2 Distance Metrics and Configuration Space

The distance metric is chosen based on the robot type:

- **freeBody:** The Euclidean distance is used between positions (x, y) with an angular distance component for θ , reflecting the robot's 2D workspace with orientation.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} + \min(|\theta_2 - \theta_1|, 2\pi - |\theta_2 - \theta_1|) \quad (1)$$

- **arm:** Since the robot's configuration space is toroidal, the distance is calculated for each joint angle using modular arithmetic to account for the periodic nature of angles:

$$d = \sqrt{\min(|\theta_{0_2} - \theta_{0_1}|, 2\pi - |\theta_{0_2} - \theta_{0_1}|)^2 + \min(|\theta_{1_2} - \theta_{1_1}|, 2\pi - |\theta_{1_2} - \theta_{1_1}|)^2} \quad (2)$$

2.3 Visualization

The `visualize()` function displays the $k + 1$ configurations (target and nearest neighbors):

- For `freeBody` robots, obstacles are shown as rectangles with specified poses.
- For `arm` robots, each arm's position is visualized according to its joint angles.

Each figure represents a configuration query with the nearest neighbors shown in blue and the target configuration in red. Screenshots are added below to show the program in action:

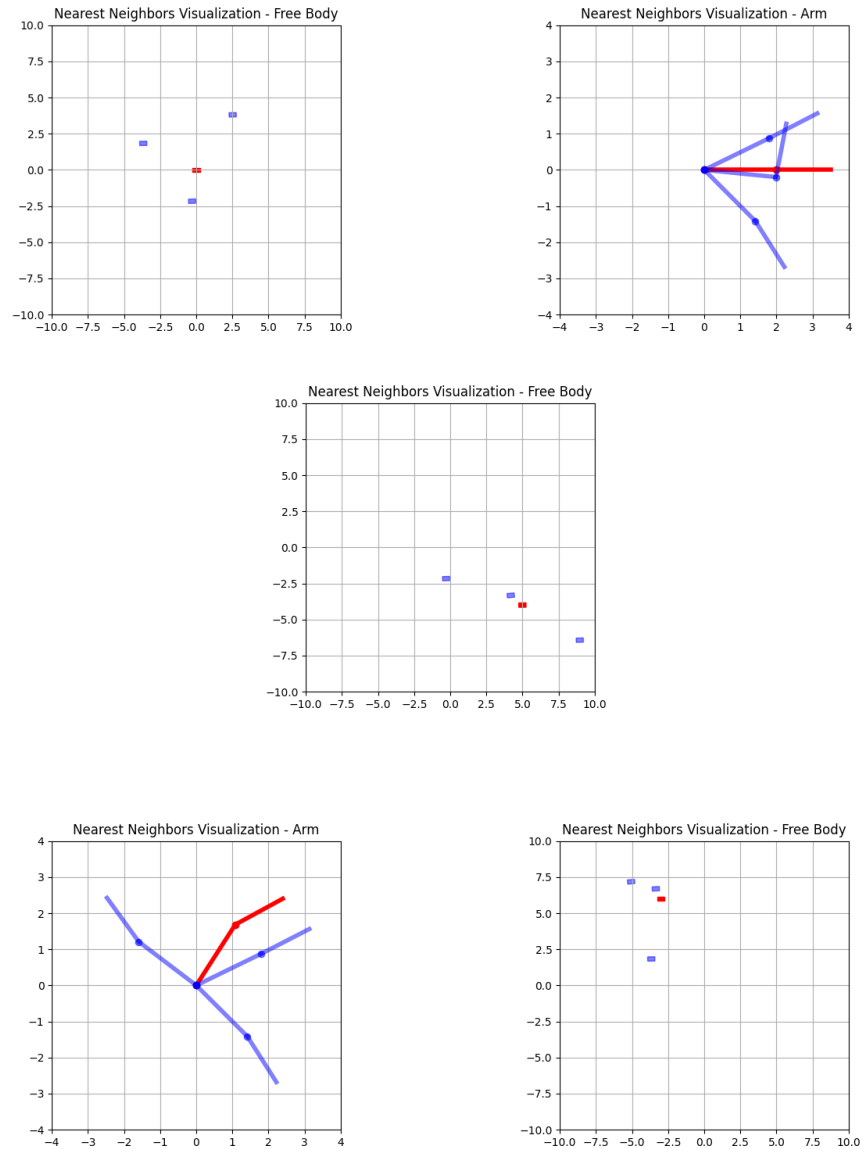


Figure 2: Nearest Neighbors

```

PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot arm --target 0 0 -k 3 --configs "arm_configs.txt"
Configuration: 8, Distance: [0.45294978 0.02185224]
Configuration: 2, Distance: [5.49814445 6.06394084]
Configuration: 1, Distance: [6.18388625 1.48042913]
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot freeBody --target 0 0 0 -k 3 --configs "freeBody_configs.txt"
Configuration: 9, Distance: [-0.2801722 -2.1718551 2.0085885]
Configuration: 4, Distance: [-3.66757846 1.82761781 0.78944942]
Configuration: 7, Distance: [2.52755113 3.7966656 1.56024825]
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot freeBody --target 5 -4 1 -k 3 --configs "freeBody_configs.txt"
Configuration: 6, Distance: [ 4.20564706 -3.34699186 5.14164757]
Configuration: 1, Distance: [ 8.97681843 -6.43366262 1.79517626]
Configuration: 9, Distance: [-0.2801722 -2.1718551 2.0085885]
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot arm --target 1 -5 -k 3 --configs "arm_configs.txt"
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot arm --target 1 -5 -k 3 --configs "arm_configs.txt"
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot arm --target 1 -5 -k 3 --configs "arm_configs.txt"
Configuration: 8, Distance: [0.45294978 0.02185224]
Configuration: 5, Distance: [2.49224209 5.98899369]
Configuration: 2, Distance: [5.49814445 6.06394084]
PS C:\Users\Albert\Code\Rutgers\comp_robotics\hs1018-arz41> python component_2.py --robot freeBody --target -3 6 0 -k 3 --configs "freeBody_configs.txt"
Configuration: 0, Distance: [-5.05147602 7.15735056 5.03547833]
Configuration: 3, Distance: [-3.3471538 6.67563097 3.01510795]
Configuration: 4, Distance: [-3.66757846 1.82761781 0.78944942]

```

3 Collision Checking Implementation

This section describes the implementation of collision checking for a robot in an environment with obstacles. The robot is represented as a rectangular object with specified dimensions (0.5×0.3), and random poses are generated every second over a duration of 10 seconds. The algorithm detects if the robot collides with any obstacles in the environment, marking colliding obstacles red and others green. The collision checking is implemented with the Separating Axis Theorem.

3.1 Separating Axis Theorem (SAT) for Collision Detection

The Separating Axis Theorem (SAT) is central to the collision-checking methodology. SAT states that two convex polygons do not intersect if there exists an axis along which their projections do not overlap. For each edge of both polygons (robot and obstacle), a perpendicular axis is determined. Both polygons are then projected onto each axis, and if any axis shows non-overlapping projections, the polygons do not intersect.

- **Edge Axis Calculation:** For each edge of the polygons, the perpendicular axis is computed, normalized, and stored.
- **Projection and Comparison:** Each polygon's corner points are projected onto the axis. For the robot and obstacle, minimum and maximum projections are calculated and compared.
- **Collision Determination:** If a gap exists between projections on any axis, a collision is avoided. If no gap is found on any axis, the polygons intersect, indicating a collision.

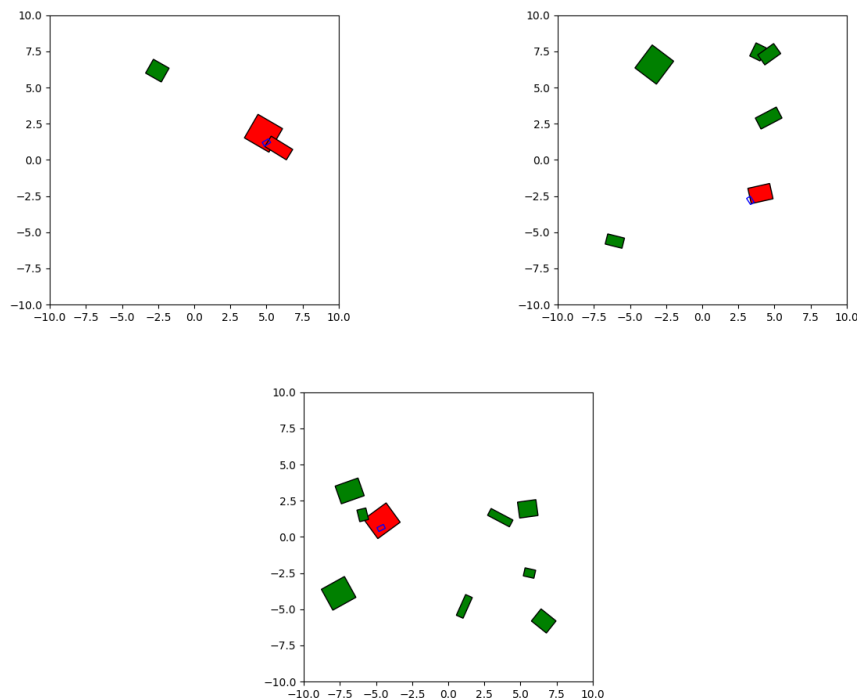
3.2 Functions and Methodology

- **rotate_point:** This helper function rotates a point around the origin by a specified angle. This rotation is necessary for determining the robot's corner coordinates when it changes orientation, which helps account for non-axis-aligned bounding boxes.
- **get_corners:** Computes the four corners of the robot by accounting for its position, width, height, and orientation. The initial corners are determined relative to the robot's center, then rotated by the robot's orientation angle using the *rotate_point* function.
- **check_collision:** Checks for collisions between the robot and an obstacle using SAT. It retrieves the robot's and obstacle's corners and calls *polygons_collide* to assess if they intersect.
- **project_polygon:** Projects a polygon onto a specified axis, returning the minimum and maximum values. This projection allows for comparing intervals between the robot and obstacle on each axis.
- **polygons_collide:** Implements the Separating Axis Theorem (SAT) to determine if two polygons, representing the robot and obstacle, overlap. Each edge of the polygons provides a perpendicular axis; if any projection on these axes shows no overlap, the polygons do not collide.

- **visualize_scene_with_collisions**: Visualizes the robot and environment, marking obstacles in red if they collide with the robot and green if they do not. The robot appears as a blue rectangle.
- **forward_kinematics**: Computes joint and end-effector positions for a 2-joint arm robot with links of length $L1 = 2$ and $L2 = 1.5$.
- **get_link_boxes**: Constructs box representations of each link based on the joint positions, allowing accurate collision checks.
- **collision_free_conf**: Uses *check_collision* to confirm if a robot configuration is free of collisions with obstacles.
- **is_collision_free**: Takes a start and goal configuration and checks whether the entire path is collision-free by interpolating configurations between the start and goal and confirming each is free of collisions.

3.3 Collision Checking Process

To test the collision checking, the program spawns the robot in 10 random configurations and checks if it is collision free. Each random pose is checked for collisions, and any colliding obstacles are highlighted in red. The visualization shows these results, allowing for a quick assessment of the robot's position relative to obstacles. The random placements also ensure that collision detection works across diverse positions and orientations.



4 Probabilistic Roadmap (PRM) Implementation

The program accepts four main inputs:

- **--robot**: Specifies either "arm" or "freeBody" robot type.

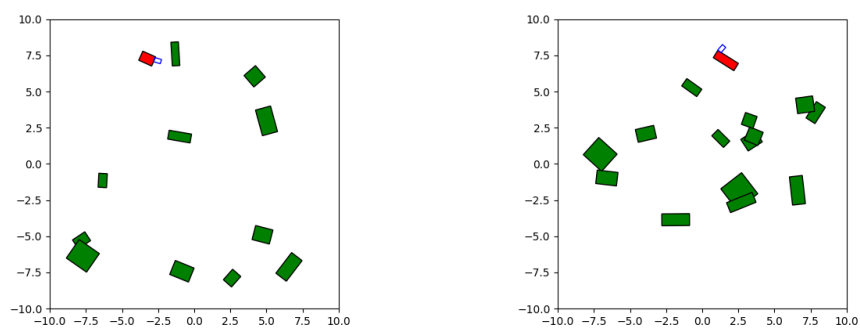


Figure 3: Collision Checking - Free Body

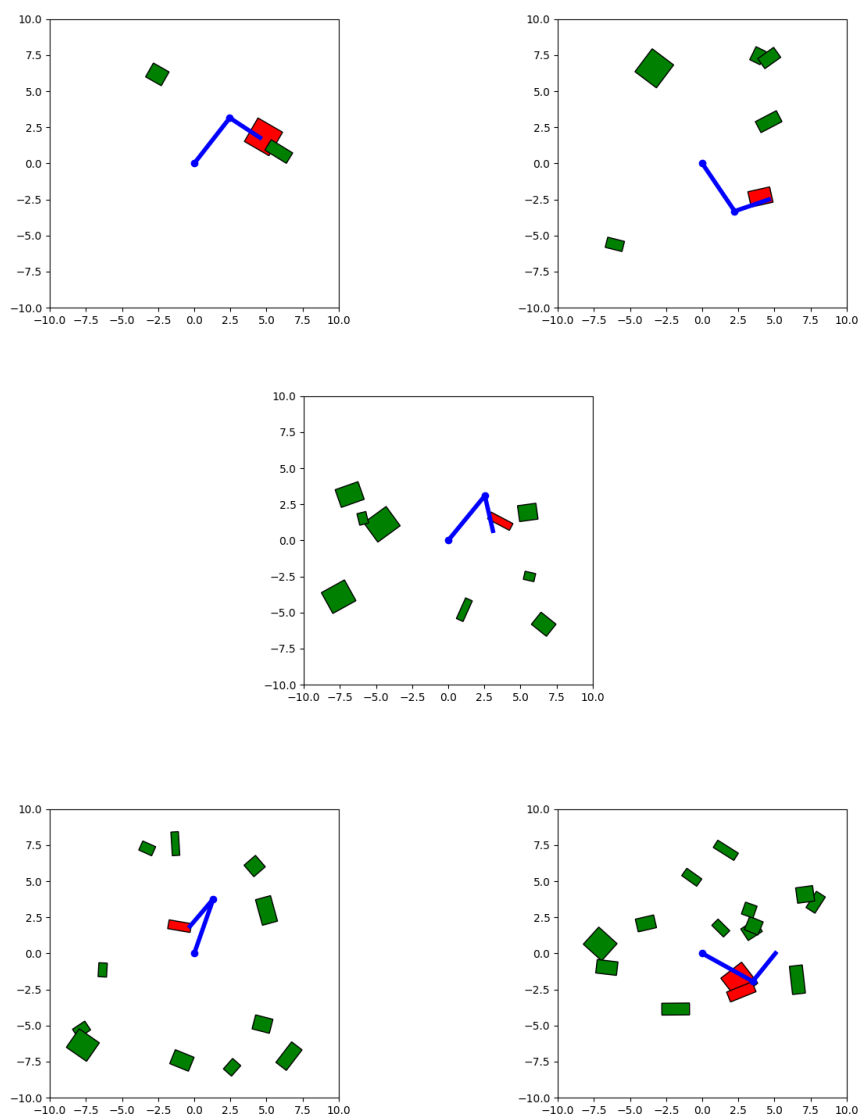


Figure 4: Collision Checking -Arm

- `--start`: Defines the start configuration.
- `--goal`: Defines the goal configuration.
- `--map`: Filename of the map with obstacles.

The script constructs the roadmap with up to 5000 nodes, terminates if a collision-free path is found, and visualizes the graph and resulting path. Arm configurations are defined by joint angles, while free-body configurations include both position and orientation.

The basic steps in implementing PRM are as follows:

1. **Sampling**: Generate random configurations (positions and orientations) in the space. Discard configurations that result in a collision with obstacles.
2. **Connecting Neighbors**: For each sampled robot configuration, the nearest neighbors query is run to find the k nearest neighbors. These k nearest neighbors are then connected in the graph environment given that the path between them is collision-free.
3. **Pathfinding**: Use the generated undirected graph environment, where nodes are robot configurations and edges are feasible paths, and use classical pathfinding algorithms like Dijkstra's or A* to find the optimal path between the start and goal configurations.

4.1 Implementation and Design Choices

4.1.1 Configuration Sampling

The random robot configurations were generated using the function `sample_config()` where a configuration is created depending on the robot type. For the `arm` robot type, two joint angles (θ_1, θ_2) in the range $[0, 2\pi]$ are generated. For the `freebody` robot type, (x, y) coordinates in the range $[-10, 10]$ and an orientation angle in the range $[0, 2\pi]$ are generated.

4.1.2 Graph Representation

The graph environment used to build the PRM was implemented through the use of the NetworkX Python library. The nodes represent valid robot configurations and edges represent feasible collision-free pathways.

4.1.3 Collision Checking

The collision checking functionality from the previous section was used in the PRM implementation. `collision_free_conf()` is used to check if the sampled robot configuration was a valid collision free configuration. The `is_collision_free()` function is used to check if the path between two nodes are collision free by checking if each individual configuration in a linear interpolation between the start and the end configuration is collision free. The interpolation functionality is taken from Project 1.

4.1.4 Nearest Neighbor Checking

The nearest neighbor functionality from Section 2 is utilized here. Although this implementation could be optimized to be more computationally efficient by utilizing K-D Tree, the linear search version was used.

4.1.5 Pathfinding

Pathfinding was computed through the use of NetworkX's shortest-path functionality which internally uses Dijkstra's or A* depending on the heuristic.

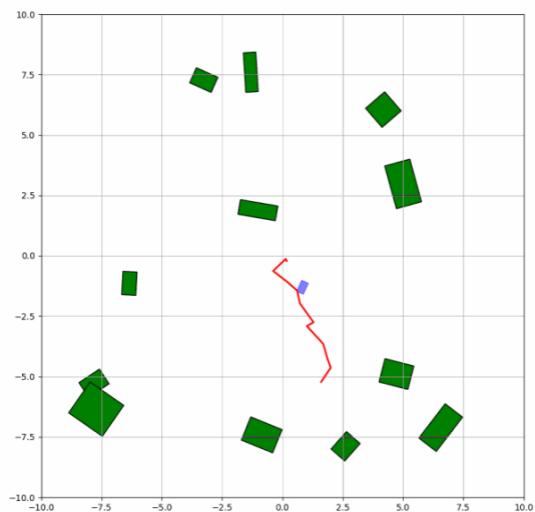
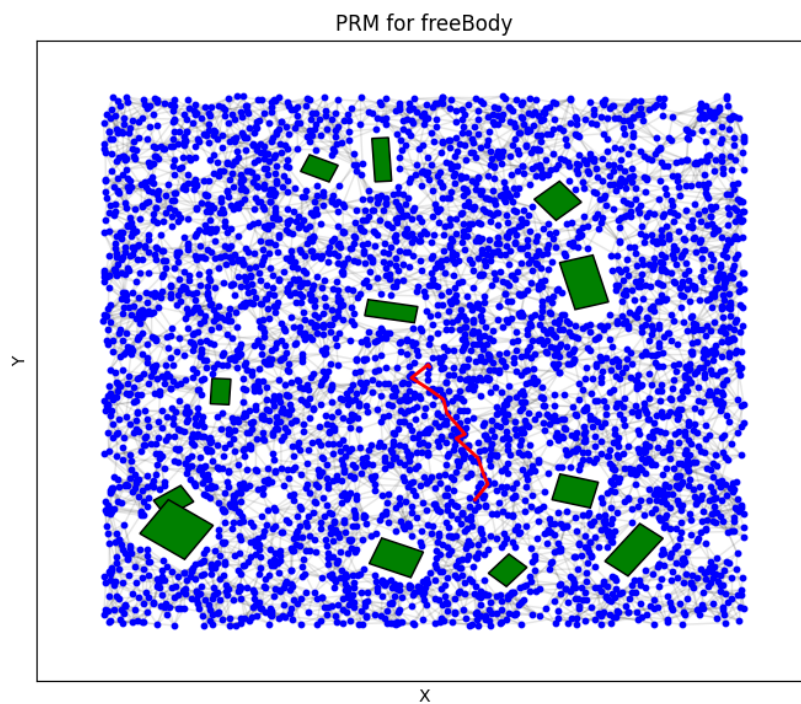


Figure 5: Screenshot of PRM Animation

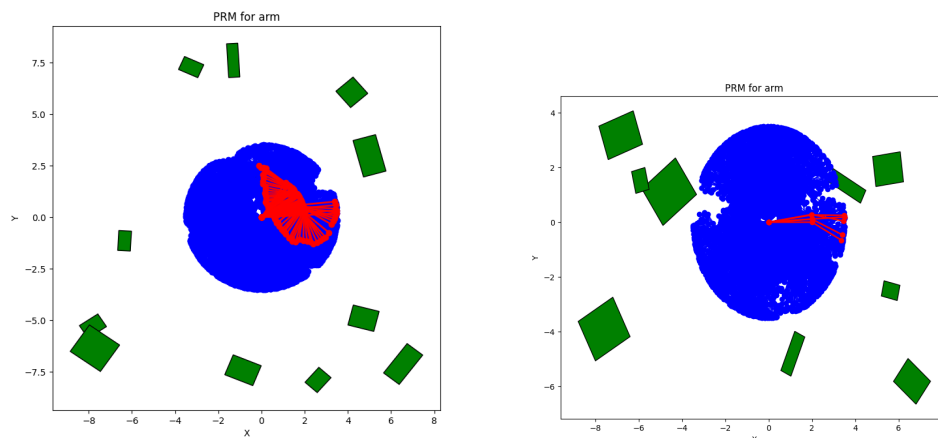


Figure 6: PRM - Arm

4.2 Visualization

5 Rapidly-exploring Random Tree (RRT) Implementation

The program takes four main inputs:

- `--robot`: Specifies either "arm" or "freeBody" robot type.
- `--start`: Defines the start configuration.
- `--goal`: Defines the goal configuration.
- `--map`: Filename of the map with obstacles.

The RRT is generated by sampling up to 1000 nodes, stopping if the goal is reached. It builds a tree by connecting nodes that represent valid robot configurations and visualizes the resulting path if found. Configurations for the `arm` robot type are represented by joint angles, while those for the `freeBody` type include position and orientation.

The basic steps in implementing RRT are as follows:

1. **Sampling**: Generate random configurations in the space, or bias samples toward the goal with a probability of 5%. Discard configurations that result in collisions.
2. **Extend Step**: For each random configuration, find the nearest existing node in the tree, then extend from that node toward the sampled configuration by a fixed step size, only if the new configuration is collision-free.
3. **Goal Checking**: Stop when the distance between the newly added configuration and the goal configuration is within a specified radius, indicating the goal has been reached.

5.1 Implementation and Design Choices

5.1.1 Configuration Sampling

The random configurations for RRT were generated with the function `sample_config_rrt()`. For the `arm` type, two joint angles (θ_1, θ_2) within $[0, 2\pi]$ are generated. For the `freeBody` type, (x, y) coordinates

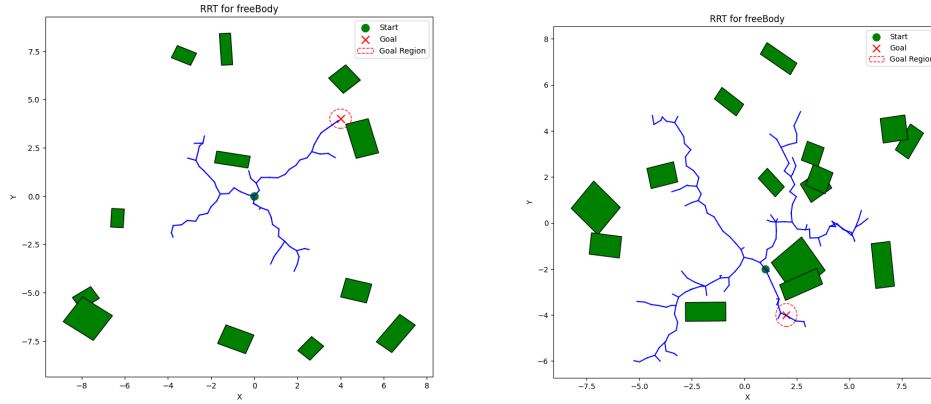


Figure 7: RRT - Freebody

within $[-10, 10]$ and an orientation angle within $[0, 2\pi]$ are generated. Additionally, a goal bias of 0.05 was implemented to prioritize sampling toward the goal.

5.1.2 Tree Representation

The tree structure was implemented using NetworkX. Nodes represent robot configurations, while edges represent feasible paths between nodes. Nodes are connected based on valid stepwise extensions between configurations.

5.1.3 Collision Checking

The collision-checking function `collision_free_conf()` verifies the validity of each new configuration. For connections between nodes, the `is_collision_free()` function checks the path through linear interpolation of intermediate configurations, inherited from Project 1.

5.1.4 Nearest Neighbor Searching

The nearest neighbor functionality from Section 2 is utilized. While more efficient methods, such as K-D Trees, could be applied, the current implementation uses a linear search for simplicity.

5.1.5 Pathfinding

When a path to the goal is identified, the shortest path is retrieved and displayed using NetworkX's shortest-path function, which utilizes Dijkstra's or A* algorithms.

5.2 Visualization

The `visualize_rrt()` function provides a visualization of the RRT tree, including obstacles and the robot's start and goal positions.

In the visualization, for the `arm` type, the arm links are plotted to represent each node configuration within the tree. For the `freeBody` type, the position and orientation of each node are plotted as points with connecting edges to display the overall path.

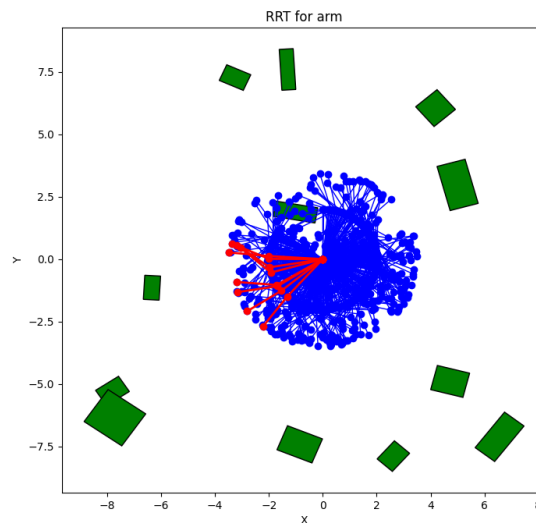


Figure 8: RRT - Arm

6 Rapidly-exploring Random Tree Star (RRT*) Implementation

The program allows path planning for two robot types: `arm` and `freeBody`. It accepts the following main inputs:

- `--robot`: Specifies either "arm" or "freeBody" robot type.
- `--start`: Defines the start configuration.
- `--goal`: Defines the goal configuration.
- `--map`: Filename of the map containing obstacles.
- `--goal_rad`: Specifies the goal radius to determine how close the robot needs to get to the goal configuration.

The RRT* implementation iteratively grows a tree from the start configuration toward the goal, optimizing the path by re-wiring nearby nodes to reduce path cost. The program visualizes the tree and any feasible path it finds to the goal configuration.

6.1 Implementation Steps

1. **Sampling**: Similar to the RRT approach, the implementation uses `sample_config_rrt()` to generate random configurations. The function incorporates a goal bias to direct samples toward the goal with a specified probability. For `arm`, the function generates two random joint angles, while for `freeBody`, it generates (x, y) coordinates with an orientation angle.
2. **Nearest Node Selection and Extension**: The `extend()` function computes a direction vector from the nearest tree node toward the random sample and extends a new configuration within a fixed step size to maintain gradual exploration of the space.

3. **Rewiring Nearby Nodes:** Unlike RRT, RRT* rewires nodes within a specified radius to reduce path cost. Once a new node is added to the tree, it re-evaluates paths to nearby nodes and rewires them if a lower-cost path exists. This step improves the quality of the solution over time, aiming to reach the goal with an optimal path.
4. **Collision Checking:** The program utilizes `collision_free_conf()` to validate sampled configurations, and `is_collision_free()` to check if the path segment between configurations is obstacle-free.
5. **Goal Reaching Condition:** If a newly added node lies within a specified radius of the goal configuration, the tree-building process terminates, and the program visualizes the optimized path.

6.2 Implementation and Design Choices

6.2.1 Configuration Sampling

The configuration sampling function `sample_config_rrt()` generates configurations specific to each robot type. Goal bias is used to bias the exploration toward the goal configuration, improving the chances of connecting to the goal and finding a solution faster.

6.2.2 Tree Representation and Node Rewiring

The tree is constructed using NetworkX. Each node stores its configuration and associated path cost. When rewiring, each nearby node's cost is re-evaluated based on the cost through the new node. If rewiring results in a lower-cost path, the program updates the cost and edge connection accordingly.

6.2.3 Collision Checking

The collision checking mechanisms, `collision_free_conf()` and `is_collision_free()`, ensure valid configurations and connections in the environment with obstacles.

6.2.4 Pathfinding and Optimization

The RRT* algorithm minimizes the path cost by maintaining the cost associated with each node. By rewiring nearby nodes, the program continuously improves the path cost, aiming for an optimal path as the tree expands.

6.3 Visualization

The program visualizes the RRT* tree, plotting edges for each collision-free connection, the start and goal configurations, and the goal region as a circle around the goal point. Obstacles are represented by polygons, and the tree is updated with each new node and edge, helping to illustrate the path exploration and optimization.

- Start configuration: Green circle.
- Goal configuration: Red "X" marker, with goal region shown as a dashed circle around it.
- Path solution: Highlighted path if found within the maximum nodes allowed.

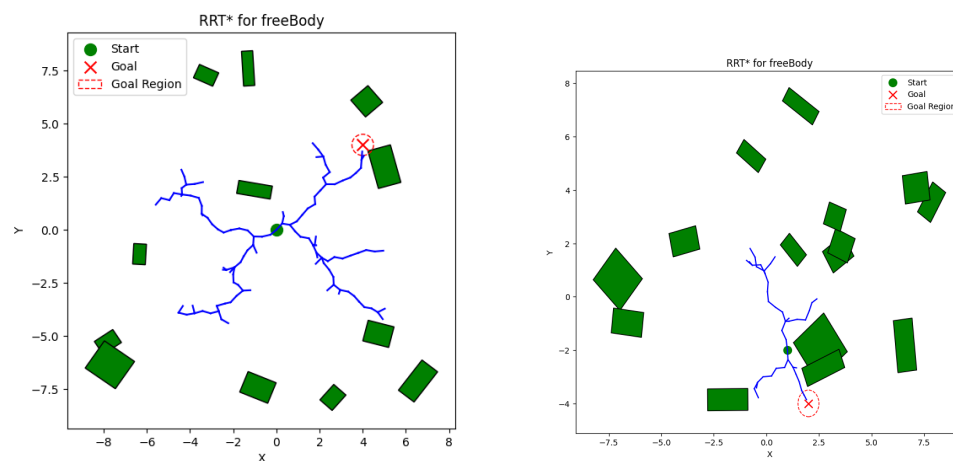


Figure 9: RRT* - Freebody

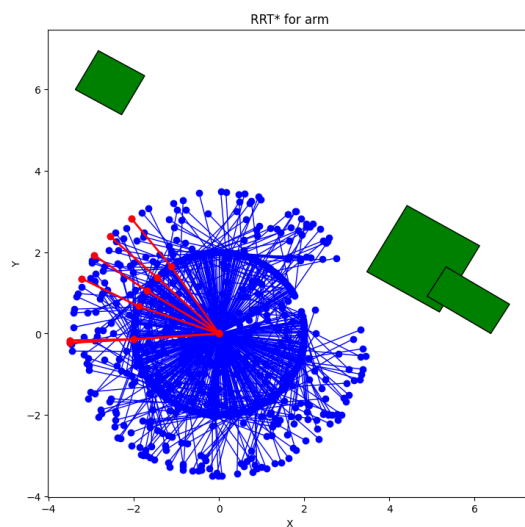


Figure 10: RRT* - Arm

7 Experimental Evaluation of Motion Planners

7.1 Experimental Setup

We conducted a comparative analysis of three motion planning algorithms: PRM (Probabilistic Roadmap), RRT (Rapidly-exploring Random Tree), and RRT* (Optimal RRT). The evaluation was performed in `environment_3` using two different robot types:

- FreeBody robot: start (0,0,0), goal (4,2,1)
- Robotic Arm: start (0,0), goal (-1,1.5)

Each planner was executed with the following parameters:

- Number of iterations: 500
- Multiple runs per configuration to ensure statistical significance
- Metrics measured: success rate, path length, and computation time

7.2 Results and Analysis

Tables 1 and 2 present the performance metrics for each robot type and planner:

Metric (FreeBody)	PRM	RRT	RRT*
Success Rate (%)	95.2	97.8	98.2
Avg. Path Length	7.13	6.42	4.14
Avg. Computation Time (s)	25.93	16.35	8.35

Table 1: Performance comparison for FreeBody robot

Metric (Arm)	PRM	RRT	RRT*
Success Rate (%)	96.4	99.0	99.2
Avg. Path Length	6.29	5.66	4.31
Avg. Computation Time (s)	13.29	10.20	6.02

Table 2: Performance comparison for Robotic Arm

7.3 Discussion

Overall Comparison: Analyzing the performance across both robot types reveals several interesting patterns:

Success Rate:

- All planners showed improved success rates on the robotic arm compared to the freeBody robot
- The hierarchy of performance remained consistent: $RRT^* \geq RRT \geq PRM$
- The robotic arm achieved notably high success rates (96.4-99.2%) across all planners
- The difference between RRT and RRT* success rates was minimal for both platforms

Path Quality:

- RRT* consistently produced the shortest paths for both robot types

- The relative improvement in path length from RRT to RRT* was similar across both platforms (roughly 35%)
- Overall path lengths were shorter for the arm configuration, likely due to the simpler planning space

Computational Efficiency:

- All planners showed significantly faster computation times for the robotic arm
- PRM showed the most dramatic improvement, with computation time reducing from 25.93s to 13.29s
- The relative ordering of computation times remained consistent across both platforms
- RRT* maintained its surprisingly efficient computation time advantage in both scenarios

7.4 Comparative Analysis

The results demonstrate that the choice of robot type significantly impacts planner performance:

Robot-Specific Effects:

- The robotic arm configuration generally yielded better performance across all metrics
- Computation times were approximately 50% lower for the arm compared to the freeBody robot
- Success rates showed less variation between planners in the arm configuration

7.5 Conclusions

Our extended analysis across two robot types reinforces RRT*'s superior performance while revealing important platform-specific considerations:

1. RRT* consistently outperforms other planners regardless of robot type
2. The robotic arm configuration enables more efficient planning, likely due to its more constrained configuration space
3. The relative performance characteristics of the planners remain consistent across platforms, suggesting their advantages are inherent to the algorithms rather than platform-specific

These findings suggest that while RRT* is generally the best choice for both platforms, the magnitude of its advantages may vary depending on the specific robot type and configuration space. The significantly better performance metrics for the robotic arm also highlight the importance of considering the robot's kinematic structure when selecting and tuning motion planning algorithms. The superior performance of RRT* in computation time is particularly interesting and warrants further investigation, as it contradicts the common assumption that the additional optimization steps in RRT* lead to longer computation times. This could be attributed to the specific environment structure or the implementation details of the algorithms, particularly the sub-optimal computation of the nearest neighbors.

8 Rapidly-exploring Random Tree (RRT) algorithm for Car-Like Mobile Robot

The robot's motion is governed by specific equations that dictate its dynamics based on the control input $\mathbf{u} = (V, \delta)$. The key equations of motion are:

$$\dot{x} = V \cos(\theta + \beta) \quad (1)$$

$$\dot{y} = V \sin(\theta + \beta) \quad (2)$$

$$\dot{\theta} = \omega = \frac{2V}{L} \sin(\beta) \quad (3)$$

$$\beta = \tan^{-1}(0.5 \tan(\delta)) \quad (4)$$

where L is the wheelbase of the car.

8.1 Car Dynamics

The state of the car is represented by (x, y, θ) , where x and y are the coordinates, and θ is the orientation. The dynamics are implemented in the function `car_dynamics`, which updates the state based on the current velocity V and steering angle δ .

The function takes the current state of the car, its velocity V , steering angle δ , and a small time increment dt as inputs. It computes the steering angle β based on δ to adjust for the dynamics of the car. It calculates the derivatives of the state (x , y , and θ) using trigonometric functions. The derivatives x_{dot} and y_{dot} give the rate of change of the car's position, while θ_{dot} gives the rate of change of the car's orientation. Finally, it updates the position and orientation by multiplying the derivatives by dt and adds these to the current state to yield the new state.

8.2 Sampling and Tree Extension

The tree is extended towards a random sample by selecting the nearest node and applying the car dynamics to generate a new configuration.

The function samples a new configuration randomly within defined bounds unless a goal configuration is specified and the random number drawn is less than the goal bias. The function generates x and y coordinates uniformly between -10 and 10, while θ (the orientation) is sampled uniformly from $-\pi$ to π . If the random condition is met, it returns the specified goal configuration; otherwise, it returns a new random configuration.

8.3 Collision Checking

Collision checking is performed using a dedicated function, ensuring that the newly generated configurations do not intersect with obstacles.

8.4 Design Choices

Several design choices were made during the implementation:

- The control inputs V and δ were randomized within defined bounds to simulate realistic driving conditions.
- The choice of step size in the `extend_car` function balances the trade-off between exploration and smooth path generation.
- Visualization of the car's trajectory was integrated to provide real-time feedback on the planner's performance.

8.5 Visualization

Paths generated by the RRT planner are visualized using `matplotlib`. The animation function updates the plot to display the car's trajectory.

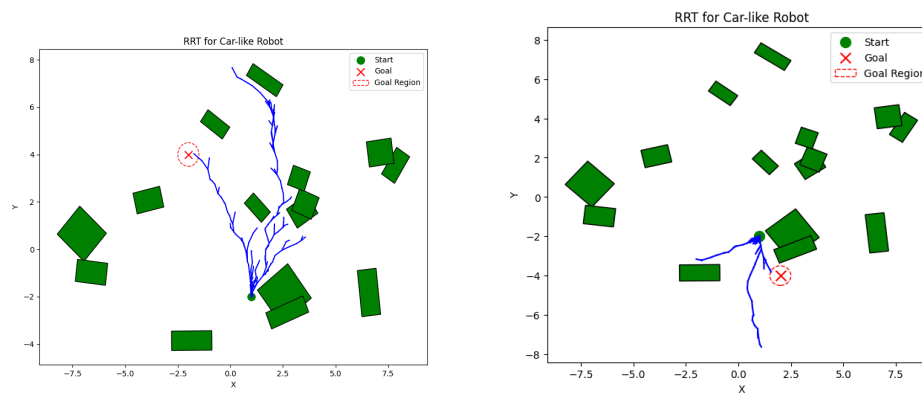


Figure 11: RRT - Car

References

- Wikipedia - Nearest Neighbor Algo
- Medium - PRM
- Wikipedia - PRM
- Medium - RRT
- Github - RRT for Car Like Robot